Shubhankar

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Human-centered AI researcher bridging machine learning and HCI, with experience in experimental evaluation, UX and Engineering.

Education

Ecoles des Mines Paris - PSL — France
Masters in High Performance Computing and Artificial Intelligence — *Oct. 2024 – Dec. 2025*

Universidad Politécnica de Madrid & Université Paris-Saclay — Madrid, Spain & Paris, France
Masters in Computer Science: Human Computer Interaction; GPA: 8.8/10 — *2022 – 2024*

Indraprastha Institute of Information Technology Delhi (IIIT-Delhi) — New Delhi, India
Bachelor of Engineering in Computer Science and Technology with senior certificate at University of Florida — *Aug. 2012 – Aug. 2016*

Professional Experience

Smart Trade — Aix-En Provence, France
Research Engineer — *Oct. 2024 – Sep. 2025*

- Developed AI-driven performance-optimization algorithms for Cockpit, a real-time trading platform used by leading global banks and corporations, under the guidance of Architect Mathieu Estrata and Prof. Youssef Mesri.

Inria, Ex-Situ Lab — Paris-Saclay, France
Research Intern — *Mar. 2024 – Aug. 2024*

- Explored Algebraic Machine Learning (AML) for music creation under Prof. Janin Koch and Prof. Wendy E. Mackay.
- Developed an interactive prototype enhancing AI explainability in HCI for professional music practice.

Samsung Research Institute — Noida, India
Software Engineer - Android Applications, Make For India - Innovation, Patents, Research, Machine Learning — *Aug. 2016 – Aug. 2022*

- **Core Samsung Android Applications**: Led development and maintenance of core Android applications, implementing new features and Proof of Concept designs based on user feedback, increasing user base by 100%.
- **United Nations Global Goals App**: Spearheaded development and launch of the United Nations Global Goals Android app and Smart Watch interface for Galaxy Watch 4, launched globally with localized content and UPI Payment.
- **Ultra Data Saving (UDS)**: Implemented UDS in Samsung Max VPN, achieving up to 50% data savings and a 23% reduction in power consumption.
- **Samsung Galaxy Store Optimization**: Optimized search functionality, doubling active users and contributing $3 million to revenue.
- **Smartcrop Architecture**: Designed and implemented Smartcrop, a computer vision-based screenshot analyzer, improving operation time by 3 seconds on millions of devices.

Core Competencies

Languages: English (Native), French (Professional), Spanish(Professional)
Programming Languages: Java, Python, C/C++, JavaScript, SQL, Kotlin, C#,
Design Tools: Adobe XD, Sketch, Figma, InVision, Marvel, Balsamiq, Illustrator, Photoshop
Frameworks and Libraries: TensorFlow, PyTorch, AngularJS, Numpy, Pandas
Tools: Git, Gerrit, Jenkins, AWS, Docker, Kubernetes
Methodologies: UX Research, Qualitative & Quantitative Methods, User-Centered Design, A/B Testing, Accessibility Design
Software Engineering: Agile Software Development Process, Object Oriented Design, Scrum, System Design

Selected Projects

Tune-Crafter | *JavaScript, Node.js, TensorFlow* — Sept. 2023 – Dec. 2023

- Developed a web application for crafting tunes using modern UI toolkits and Google Media Pipe.
- Tracks your fingers using Computer Vision and map musical instruments to gestures with Explainable AI

AI for Computer Games: Horror Video Game | *Unity3D, C#* — Jan. 2016 – Apr. 2016

- Implemented an A* algorithm in a horror-themed video game, enhancing AI pathfinding and user experience, leading to an immersive user experience.

Research Method for Human-Centered Computing | *Python* — Jan. 2016 – Apr. 2016

- Evaluated driver satisfaction with autonomous braking in simulated driving conditions for BMW Cars, leading to design recommendations adopted by the company.

Newspaper | *Angular, HTML, CSS, TypeScript, Android, Java* — Apr. 2023 – Jun. 2023

- Developed an online newspaper application for cross-platform support with user authentication, increasing user engagement by 20%.
- Conducted usability testing with 50+ users to ensure a seamless experience across platforms, resulting in a 15% reduction in user-reported issues.

HPC–AI & HCI Projects (Mines Paris – PSL)

Parallel Programming for Scientific Computing | *C++, MPI, OpenMP* Oct. 2024 – Mar. 2025

- Implemented parallel algorithms for dense/sparse matrix–vector multiplication and QR decomposition on distributed-memory systems.
- Structured MPI communicators and collective operations to reduce synchronization cost; profiled computation/communication overlap on multi-node clusters.
- Analyzed strong/weak scaling from 2 to 32 processes and documented memory-bandwidth and latency effects on large matrices.

Large-Scale Machine Learning with Spark | *Python, PySpark, Spark MLlib* Nov. 2024 – Feb. 2025

- Built a distributed image-classification pipeline: median filtering for denoising, feature extraction, and MLlib model training/evaluation on cluster datasets.
- Integrated PySpark DataFrames with pipeline APIs for reproducible preprocessing and model selection under distributed execution.
- Tuned partitioning and shuffle behavior; reported task skew and I/O characteristics to guide cluster configuration.

Distributed Algebra & Conjugate Gradient Optimization | *C++, MPI, BLAS/LAPACK* Oct. 2024 – Feb. 2025

- Implemented Conjugate Gradient solvers with communication-hiding/avoiding strategies to mitigate global-reduction bottlenecks.
- Compared baseline vs. optimized iterations across nodes; visualized residual norms and per-iteration timings to characterize convergence and latency.
- Documented numerical stability trade-offs and precision impacts on large sparse systems.

Convex Optimization Algorithms from Scratch | *Python, NumPy, Matplotlib* Dec. 2024 – Mar. 2025

- Implemented Gradient Descent, SGD, Momentum, RMSProp, and Adam; evaluated learning-rate schedules and convergence on convex objectives and logistic regression.
- Produced experiment reports with reproducible seeds and plots for loss curves, gradient norms, and sensitivity to hyperparameters.

GPU Acceleration of Transformer Components | *C++, CUDA, PyTorch* Jan. 2025 – Mar. 2025

- Ported core Transformer blocks (multi-head attention, layer norm, positional encodings) into custom CUDA kernels to study kernel fusion and memory access patterns.
- Benchmarked kernel execution vs. PyTorch reference on small GPT-style models; examined shared-memory usage, occupancy, and warp-level parallelism.

CPU Multithreading & Vectorization | *C, Pthreads, SIMD* Jan. 2025 – Mar. 2025

- Optimized a compute-heavy numerical routine using Pthreads for task parallelism and SIMD intrinsics for data-level parallelism.
- Used Linux `perf` and cache-miss profiling to iteratively refine scheduling, affinity, and memory layout for throughput gains.

Transformer Language Model (GPT-2 Class) from Scratch | *Python, PyTorch* Feb. 2025 – Mar. 2025

- Implemented tokenizer, positional encodings, multi-head self-attention, residuals, and autoregressive decoding to reproduce GPT-2–style behavior on toy corpora.
- Analyzed attention maps and training stability under different normalization placements and residual paths.

Positive & Unlabeled Learning for Protein Function | *Python, scikit-learn/PyTorch* Nov. 2024 – Jan. 2025

- Applied PU-learning to infer protein functionality from sequence-derived features; compared risk estimators and class-prior assumptions.
- Evaluated performance with stratified validation and reported calibration/thresholding behavior under label scarcity.

Virtual Reality Course Projects | *Unity3D, Oculus Rift, Google Cardboard, GearVR* Jan. 2016 – Apr. 2016

- At **University of Florida**: Developed four individual VR experiences (3 minutes each) centered around interactive narratives. All experiences ran on Google Cardboard, Oculus Rift, and GearVR, providing compelling immersion:
  - **Simulator Sickness Reduction**: Applied design and rendering techniques to minimize motion-to-photon latency, abrupt camera motion, and mismatched vestibular cues.
  - **Telepresence Experience**: Reconstructed the Apollo Rocket Launch Site using spatialized sound and panoramic imagery, enabling time-and-location travel in VR.
  - **VR Exhibit Visualization**: Created a virtual concert environment visualizing Michael Jackson's music using animated particle systems and synchronized lighting.
  - **Educational PTSD Scenario**: Built an interactive narrative explaining the onset of PTSD in soldiers, featuring scenario-based triggers and educational text overlays.

Research Method for Human-Centered Computing | *Python, Statistical Analysis* Jan. 2016 – Apr. 2016

- At **University of Florida**: Designed and conducted a human-subjects study evaluating driver satisfaction with autonomous braking behavior in simulated driving environments, comparing it to BMW Active Cruise Control. Tasks included: formulating hypotheses, designing experimental protocols, recruiting participants, collecting quantitative data (reaction times, stopping distances) and qualitative feedback, performing statistical analysis (t-tests, ANOVA), and producing evidence-based recommendations for automotive HCI design.

Artificial Intelligence for Computer Games | *Unity3D, C#* Jan. 2016 – Apr. 2016

- At **University of Florida**: Created a horror-themed video game integrating A* pathfinding for enemy AI that actively searches for and intercepts the player when proximity thresholds are crossed. Designed the game loop, integrated soundscapes and dynamic lighting to enhance tension, optimized AI performance to prevent frame drops, and conducted iterative play-testing to refine difficulty and user engagement.

Selected Projects (Continued)

Television Guide (Buzzset) | *Android* Jan. 2015 – Apr. 2015

- At **IIIT-Delhi**: Built an Android app displaying trending TV shows, with features for reminders and favorites. Developed through iterative prototyping, contextual inquiry, affinity diagramming, and other HCI methods.

Finger Detection & Counting | *C#, OpenCV (EmguCV)* Aug. 2014 – Nov. 2015

- At **IIIT-Delhi**: Implemented a real-time computer vision system to detect and count extended fingers using skin segmentation, contour analysis, and convex hull detection. Mapped detected counts to system-level keyboard shortcuts to control mouse pointer, media player, and presentation software hands-free.

Attack on LSB Matching Steganography | *MATLAB* Feb. 2015 – Apr. 2015

- At **IIIT-Delhi**: Reproduced and tested an ICIP research method for attacking LSB matching steganography. Implemented an algorithm to calculate alteration rate of neighborhood gray levels, trained a binary classifier to distinguish between cover and stego images, and validated performance on benchmark image datasets.

Secure File System on Interactive Shell | *C* Jan. 2015 – Mar. 2015

- At **IIIT-Delhi**: Built a secure client–server file system allowing authenticated users to create, move, and list files via pre-defined commands. Implemented Access Control Lists (ACLs) with file ownership permissions and robust input validation to prevent unauthorized operations.

Gamify Tasks | *Android, PHP, Google Cloud Messaging* Feb. 2015 – Apr. 2015

- At **IIIT-Delhi**: Developed a location-aware task management application for posting and accepting tasks within proximity radius. Implemented backend in PHP with MySQL, push notifications using Google Cloud Messaging, and gamified UI elements to increase user engagement.

Line Following Robot | *Embedded Systems* Jan. 2013 – Mar. 2013

- At **IIIT-Delhi**: Designed and assembled a dual-function robot capable of autonomous line-following using IR sensors and manual remote control via mobile phone. Implemented DTMF decoding through the 3.5mm audio jack for wireless control.

Foundations of Finance Project | *Quantitative Analysis* Oct. 2014 – Nov. 2014

- At **IIIT-Delhi**: Performed comprehensive financial analysis of the top-5 pharma companies by market cap, using ratio analysis and industry benchmarks. Calculated WACC, beta, and cost of debt for Axis Bank using historical stock data and financial statements, culminating in an investment recommendation report.

Compilers Course Project | *C, Lex, Yacc* Oct. 2014 – Nov. 2014

- At **IIIT-Delhi**: Developed a three-phase compiler: lexical analysis with Lex to generate tokens, syntactic parsing with Yacc to construct Abstract Syntax Tree, and semantic analysis with intermediate code generation for execution on a virtual machine.

NightScape Analytica | *AI, IoT, Urban Analytics* Jan. 2023 – May 2023

- At **Universidad Politécnica de Madrid (UPM)**: Concept and pitch for an AI platform that optimizes public lighting to cut energy use and light pollution while supporting local communities.
- Defined approach (predictive savings, IoT integration, Dark Sky–aligned solutions) and a savings-linked pricing model to align incentives and fund deployment at city scale.

Technology Watch: Smart Wheelchairs (EIT Digital I&E Minor) | *Tech Watch, LIDAR, Market Analysis* Jan. 2023 – May 2023

- At **UPM**: Mapped research/patent trends and R&D (e.g., CROWDBOT, H2R) across LIDAR in accessibility, vehicles, intelligent wheelchairs, and real-time perception; identified cost as the key adoption barrier.
- Concluded opportunity to integrate **LIDAR** with affordable devices (e.g., smartphones); prioritized features from user data (obstacle avoidance 100%, remote control 80%, auto-brake 67%); outlined short/long-term tech roadmap and target markets (individuals, hospitals, care facilities).

Digital Nomads: Co-Work Partner & Place Finder (HCI) | *HCI, Hi-Fi Prototype, UX Research* Jan. 2023 – May 2023

- At **UPM**: Designed and evaluated a mobile prototype for matching people/places to work; conducted **16-participant** usability tests (20–31 yrs; students, freelancers, corporate workers).
- Found strong usability with **SUS = 85.5**; identified key fixes (clearer distance filter, status visibility) and efficiency issues (filter discoverability), informing next-iteration design changes.

Portfolio (in progress) — Blog (poetry, art & memes)